

IDENTIFICATION

PRODUCT CODE: MAINDEC-8E-D6CA-D
PRODUCT NAME: VC-8E DISPLAY DIAGNOSTIC
DATE CREATED: JUNE 21, 1971
MAINTAINER: DIAGNOSTIC GROUP
AUTHOR: PATRICK COYNE

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4.

STARTING PROCEDURE

CONTROL SWITCH SETTING

SWITCH REGISTER	SET AS	ACTION ON PROGRAM
0	1 0	PROCEED TO NEXT CALIBRATE BIT.
1	1 0	Y AXIS X AXIS
2	1 0	VR03A VR14
3	1 0	VR14 CHANNEL 2 VR14 CHANNEL 1
4	1 0	EXIT SCOPE LOOP HANG IN SCOPE LOOP
5	1 0	(DIAGNAL LINE TEST) PLOT UL TO LR DIAGONAL PLOT LL TO UR DIAGONAL (VERTICAL OR HORIZONTAL BAR TEST)
5	1 0	HALT LINE MOVEMENT CONTINUE LINE MOVEMENT
6	1 0	SELECT 615X IOT SELECT 605X IOT
7	1 0	PERFORM TEST SELECTED BY SWITCHES 8-11. RETURN/STAY IN DISPATCH ROUTINE.
8	CONTAINS NUMBER OF TEST	
9	TO BE EXECUTED.	
10	(REFER TO TEST SELECTION TABLE)	
11		

VC-8E DISPLAY DIAGNOSTIC

1. ABSTRACT

THE VC-8E DISPLAY DIAGNOSTIC IS A PROGRAM WHICH FACILITATES THE CALIBRATION CHECK-OUT, AND DIAGNOSIS OF A VC-8E DISPLAY. ALL ERRORS ARE VISUAL EXCEPT FOR THE CONTROL LOGIC TEST, WHICH PROVIDES ERROR TYPEOUT AND SCOPE LOOPS.

2. REQUIREMENTS

EQUIPMENT

PDP-8E COMPUTER, TTY OR HIGH SPEED READER
M869 QUAD MODULE (DISPLAY CONTROL)
M885 QUAD MODULE (D/A CONVERTER)
TEKTRONIX 453 SCOPE OR EQUIVALENT
VR-14, VR23A OR EQUIVALENT DISPLAY

STORAGE

THE PROGRAM OCCUPIES MEMORY LOCATIONS 0000 TO 4600

3. LOADING PROCEDURE

LOAD PROGRAM VIA BINARY LOADER.

5. ERRORS

EXCEPT FOR THE CONTROL LOGIC TEST ALL ERRORS ARE DETERMINED BY VISUAL INSPECTION.

AN ERROR OCCURRING DURING THE CONTROL LOGIC TEST WILL CAUSE AN ERROR TYPEOUT GIVING THE NUMBER OF THE TEST AND AN IDENTIFICATION OF THE ERROR. THE PROGRAM WILL THEN ENTER A SCOPE LOOP, UNLESS SW4 HAS PREVIOUSLY BEEN SET TO A ONE, TO EXIT THE SCOPE LOOP SIMPLY PUT SW4 TO A ONE.

6. RESTRICTIONS

STARTING RESTRICTIONS

NOEN, PROGRAM MAY BE RESTARTED FROM LOCATION 200 AT ANY TIME.

7. PROGRAM DESCRIPTION

THE MASTER DISPATCH ROUTINE FOR TRANSFERRING CONTROL TO A SPECIFIC TEST STARTS AT LOCATION 200. TO SELECT A PARTICULAR TEST, SET SW8-11 TO THE TEST NUMBER DESIRED AND THEN SET SW7 TO A ONE. RECOVERY TO THE DISPATCH ROUTINE CAN BE MADE FROM ANY TEST BY SETTING SW7 TO A ZERO.

IN ANY TEST WHICH THE OPERATOR HAS THE OPTION OF SELECTING THE X OR Y AXIS, IT IS NECESSARY TO DO SO BEFORE THE TEST IS ENTERED. WHEN THE OPERATOR DECIDES TO CHANGE THE AXIS IT IS NECESSARY TO EXIT THE TEST BY PUTTING SW7 TO A ZERO, CHANGE THE AXIS SELECT SWITCH (SW1) TO THE APPROPRIATE POSITION AND THEN RE-ENTER THE TEST BY PUTTING SW7 TO A ONE.

IT SHOULD BE NOTED THAT TWO OF THE TESTS IN THIS DIAGNOSTIC WERE INCLUDED TO COVER A MINIMUM CONFIGURATION SYSTEM WHICH MEANS THE USER MAY NOT HAVE A DISPLAY ON THE SYSTEM. IF A DISPLAY IS NOT AVAILABLE THE "RAMP TEST" AND "DC CALIBRATION TEST" MUST BE RUN TO FACILITATE THE CHECKOUT OF THE D/A MODULE. REFER TO THE INDIVIDUAL TEST DESCRIPTION FOR MORE DETAILED INFORMATION ON EACH TEST.

ANY SYSTEM EQUIPPED WITH A VR-14 DISPLAY SHOULD BE CHECKED FOR CHANNEL SELECTION ABILITY. ANY OF THE TESTS WHICH DISPLAYS A PATTERN ON THE SCREEN CAN BE MADE TO DISPLAY THE PATTERN ON VR-14 CHANNEL ONE OR TWO UNDER CONTROL OF SW3. SW3=0, SELECTS CHANNEL 1.
SW3=1, SELECTS CHANNEL 2.

PROGRAM AND/OR OPERATOR ACTION

- A. LOAD PROGRAM INTO MEMORY PER SECTION 3.
- B. SET ADDRESS TO 200
- C. LOAD ADDRESS
- D. TEST THAT IS TO BE RUN MAY NOW BE SELECTED VIA SWITCHES 8-11. SW7 MUST BE SET TO A ONE TO PERFORM TEST. PROGRAM WILL TYPE

"SELECT TEST"

ANY TIME SW7 IS A ZERO AND WILL HANG IN DISPATCH ROUTINE UNTIL SW7 IS SET TO A ONE.

- E. THE VC-8E CAN OPERATE WITH EITHER OF TWO SETS OF IOT INSTRUCTIONS, 605X AND 615X, THROUGH THE USE OF JUMPER CONNECTIONS ON THE M869 CONTROL BOARD. REFERENCE THE ENGINEERING SPECS FOR THE CONFIGURATION OF THESE JUMPERS. THESE IOT'S CAN BE CHANGED AT ANY TIME BY THE SETTING OF SW6 (REFER TO CONTROL SWITCH SETTING TABLE). IT IS NECESSARY THAT SW6 BE PUT IN THE DESIRED POSITION BEFORE ENTERING THE DISPATCH ROUTINE THAT IS BEFORE PUTTING SW7 TO A ZERO.

- F. DEPRESS CLEAR, CONTINUE,

DISPLAY TEST SELECTION

SW8 TO 11	TEST SELECTED
0000 (0)	NO TEST
0001 (1)	CONTROL LOGIC TEST
0010 (2)	RAMP SLEWING
0011 (3)	DC CALIBRATION
0100 (4)	DISPLAYED CALIBRATION
0101 (5)	CROSSING DIAGONALS TEST
0110 (6)	HORIZONTAL FLYBACK TEST
0111 (7)	VERTICAL FLYBACK TEST
1000 (10)	CORNERS TEST
1001 (11)	DIAGONAL LINE TEST
1010 (12)	VERTICAL BAR TEST
1011 (13)	HORIZONTAL BAR TEST
1100 (14)	SINGLE POINT PLOT TEST
1101 (15)	NO TEST
1110 (16)	NO TEST
1111 (17)	NO TEST

AMP TEST

THIS TEST GENERATES A SAWTOOTH PATTERN AT THE X OR Y DAC OUTPUTS (DEPENDING ON THE SETTING OF SW1). TO OBSERVE THIS PATTERN IT IS NECESSARY TO HANG A SCOPE PROBE ON THE TEST POINTS LABELED X AND Y OF THE M885 D/A BOARD. THE WAVEFORM WILL START AT -5 VOLTS, RISE IN A RAMP TO +5 VOLTS AND DEFLECT FULL SCALE (10 VOLTS) BEFORE RISING AGAIN. THE RAMP SHOULD BE A STRAIGHT UNBROKEN LINE. ANY BREAKS OR STEPS IN THE RAMP WOULD INDICATE THAT A BIT IS NOT SWITCHING OR IS NOT WEIGHTED CORRECTLY. THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

DC CALIBRATION TEST

THIS TEST AIDS IN THE CALIBRATION OF THE X AND Y D/A'S WHEN NO DISPLAY IS AVAILABLE. SW1 IS AGAIN USED TO SELECT THE X OR Y AXIS AND SHOULD BE SET PRIOR TO ENTERING THE ROUTINE. UPON ENTERING THIS TEST THE FIRST VALUE OF THE CALIBRATION TABLE IS LOADED INTO THE DAC SELECTED, AND OUTPUT TO THE TELETYPE. THE PROGRAM WILL THEN HALT. WHEN THE OPERATOR WISHES TO CONTINUE TO THE NEXT CALIBRATION VALUE HE SIMPLY DEPRESSES KEY CONTINUE. THE OPERATOR CAN GO THROUGH THE ENTIRE CALIBRATION TABLE IN THIS MANNER. THE USER CAN EXIT THE TEST AT ANY TIME BY PUTTING SW7 TO A ZERO BEFORE HE DEPRESSES KEY CONTINUE, OR BY RESTARTING THE PROGRAM AT LOCATION 200. FOR THE VOLTAGE VALUES WHICH SHOULD BE OBSERVED FOR EACH OF THE VALUES OF THE CALIBRATION TABLE, AND FOR A STEP-BY-STEP PROCEDURE ON HOW TO IMPLEMENT THIS TEST REFER TO THE ENGINEERING SPECIFICATIONS.

DC CALIBRATION TABLE

0777
0776
0775
0773
0767
0757
0740
0737
0720
0677
0602
0577
0400
0377
0000
0777
1000

CONTROL LOGIC TEST

THIS TEST EXERCISES THE CONTROL LOGIC PORTION OF THE VC8E. IT IS DIVIDED UP INTO 12 SUB-TESTS. THIS IS THE ONLY TEST IN THIS DIAGNOSTIC WHICH CONTAINS SCOPE LOOPS AND ERROR TYPE-OUTS. WHEN AN ERROR IS ENCOUNTERED AN ERROR MESSAGE IS TYPED OUT GIVING A BRIEF DESCRIPTION OF THE ERROR AND THE PROGRAM WILL GO INTO A SCOPE LOOP ON THE ERROR. THE USER MAY EXIT THE SCOPE LOOP AT ANY TIME BY PUTTING SW4 TO A ONE. THE PROGRAM WILL THEN CONTINUE TO LOOP THROUGH THE TEST, TYPING OUT ALL ERRORS THAT ARE ENCOUNTERED BUT NOT ENTERING A SCOPE LOOP. IF SW4 IS THEN PUT BACK IN THE ZERO POSITION THE PROGRAM WILL ENTER A SCOPE LOOP ON THE NEXT ERROR IT ENCOUNTERS. THERE ARE TWO SWITCHES LOCATED ON THE CONTROL LOGIC MODULE (M869), ONE (Z) CONTROLLING THE POLARITY OF THE INTENSIFY PULSE AND ONE (DELAY) WHICH DETERMINES THE TIME DELAY IN SETTING THE DONE FLAG AFTER GIVING A LOAD X OR LOAD Y COMMAND.

THESE SWITCHES ARE SET IN THE FOLLOWING MANNER:

SWITCH NAME	POSITION	DISPLAY
Z	-	VR-14
	+	VR03A
DELAY	L	VR-14
	S	VR03A

SWITCH 2 SHOULD BE SET PRIOR TO ENTERING THE CONTROL LOGIC TEST TO SELECT EITHER A VR-14 OR VR03A MODE OF OPERATION.

SW2=0, SELECTS VR-14

SW2=1, SELECTS VR03A

THE MESSAGE "CONTROL LOGIC TEST" IS TYPED UPON ENTERING THE TEST AND AFTER EVERY COMPLETE PASS.

THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

VERTICAL FLYBACK TEST

THIS TEST IS IDENTICAL TO THE HORIZONTAL TEST EXCEPT THAT THE LINES ARE PLOTTED IN THE VERTICAL DIRECTION AT THE TOP AND BOTTOM EDGES OF THE DISPLAY.
THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

CORNERS TEST

THIS TEST COMBINES THE PREVIOUS TWO TESTS, IN THAT THROUGH THE USE OF VERTICAL AND HORIZONTAL LINE SEGMENTS CORNERS ARE FORMED IN EACH OF THE FOUR CORNERS OF THE DISPLAY. AN ADDED FEATURE IS THE USE OF INTERSECTING DIAGONAL LINE SEGMENTS IN EACH OF THE FOUR CORNERS. AGAIN ALL LINES SHOULD BE STRAIGHT AND UNBROKEN AND THE DIAGONAL LINES SHOULD INTERSECT AT THE CENTER OF EACH CORNER.
THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

DIAGONAL LINE TEST

THIS TEST DISPLAYS A FULL (1024 POINTS) DIAGONAL LINE. UPON ENTERING THE ROUTINE SW5 IS TESTED TO DETERMINE IT'S POSITION, SW5=0 WILL CAUSE A DIAGONAL LINE TO BE DISPLAYED FROM THE LOWER LEFT CORNER TO THE UPPER RIGHT CORNER OF THE SCREEN, SW5=1 WILL CAUSE A DIAGONAL LINE TO BE DISPLAYED FROM THE UPPER LEFT CORNER TO THE LOWER RIGHT CORNER OF THE SCREEN.
THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

VERTICAL BAR TEST

THIS TEST PLOTS A FULL VERTICAL BAR (1024 POINTS) MOVING HORIZONTALLY ACROSS THE DISPLAY, THE MOVEMENT OF THE BAR CAN BE CONTROLLED BY SW5, SW5=0 ALLOWS THE BAR TO MOVE ACROSS THE SCREEN, SW5=1 HALTS THE MOVEMENT OF THE BAR. THIS TEST ALLOWS THE USER TO EXAMINE THE CRT FOR SCOPE BURNS.
LIKE ALL THE TESTS IN THIS DIAGNOSTIC THIS TEST CAN BE EXITED BY PUTTING SW7 TO A ZERO, HOWEVER THIS TEST WILL ONLY EXIT AT THE COMPLETION OF A PASS OF THE BAR ACROSS THE SCREEN.
THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

DISPLAYED CALIBRATION TEST

THIS TEST IS USED TO CALIBRATE THE D/A'S WHEN A VR-14 OR EQUIVALENT DISPLAY IS AVAILABLE. UPON ENTERING THE TEST THE FIRST VALUE OF A CALIBRATION TABLE IS OUTPUT OF THE TELETYPE AND LOADED INTO THE X OR Y D/A SELECTED BY SW1. THE VALUE IS ALSO PLOTTED ON THE DISPLAY. FOR EACH CHANGE OF SW2 THE PROGRAM PROCEEDS TO THE NEXT CALIBRATION VALUE, AND THE LINE ON THE SCREEN WILL BECOME LARGER UNTIL A SOLID STRAIGHT LINE IS DISPLAYED ON THE SCREEN. AS THE LINE GROWS IT SHOULD BE OBSERVED FOR GAPS OR OVERLAPPED DOTS. EITHER OF THESE CONDITIONS WILL NECESSITATE AN ADJUSTMENT OF ONE OF THE CONTROL POTS ON THE M885 D/A MODULE. REFER TO THE ENGINEERING SPECIFICATIONS FOR THE NAMES AND LOCATIONS OF THESE POTS. THIS PROCEDURE MUST BE FOLLOWED FOR BOTH THE X AND Y AXIS.

DISPLAYED CALIBRATION TABLE

0022
0021
2023
0027
0017
0037
2077
0177
0377
2777
1777

CROSSING DIAGONALS TEST

THIS TEST DISPLAYS TWO DIAGONAL LINE SEGMENTS OF EQUAL LENGTH WHICH SHOULD CROSS IN THE CENTER OF THE SCREEN. THE LINES SHOULD BE STRAIGHT AND UNBROKEN WITH NO EVIDENCE OF ANY TRACE ON THE TRANSITION POINTS (ENDS) BETWEEN THE TWO LINES. THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

HORIZONTAL FLYBACK TEST

THIS TEST DISPLAYS FOUR HORIZONTAL LINE SEGMENTS AT THE CORNERS OF THE DISPLAY. IT IS USED TO CHECK FOR ANY SIGNS OF FLYBACK TRACES AT THE ENDS OF THE LINES AND ALSO FOR HYSTERESIS INTERFERENCE ON MAGNETIC DEFLECTION DISPLAYS. THE LINE SEGMENTS ARE PLOTTED IN THE FOLLOWING ORDER:
256 POINTS TO THE RIGHT AT THE LOWER LEFT HAND CORNER;
256 POINTS TO THE RIGHT AT THE UPPER LEFT HAND CORNER;
256 POINTS TO THE LEFT AT THE LOWER RIGHT HAND CORNER;
256 POINTS TO THE LEFT AT THE UPPER RIGHT HAND CORNER;
ALL LINE SEGMENTS SHOULD BE STRAIGHT WITH NO DISTORTION. THIS TEST IS REPEATED AS LONG AS SW7 IS A ONE, AND SW8-11 INDICATE THIS TEST.

HORIZONTAL BAR TEST

THIS TEST IS IDENTICAL TO THE VERTICAL BAR TEST EXCEPT THAT
A HORIZONTAL BAR IS MOVED IN THE VERTICAL DIRECTION.

SINGLE POINT PLOT TEST

THIS TEST DISPLAYS A POINT DETERMINED BY THE SETTING OF THE
SWITCHES. UPON SELECTION OF THIS TEST THE COMPUTER WILL
STOP TO ALLOW THE USER TO SET IN:

- A. THE "X" COORDINATE.
- B. THE "Y" COORDINATE.
- C. RESET THE SWITCH OPTIONS.

8. LISTING

7VC8E POINT PLOT DISPLAY DIAGNOSTIC

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//INSTRUCTION EQUALITIES//

4130	DICL=JMS	XDICL	/CLEAR ENABLES, FLAGS, DELAYS.
4134	DICD=JMS	XDICD	/CLEAR DONE FLAG.
4140	DISD=JMS	XDISD	/SKIP ON DONE FLAG, DO NOT CLEAR FLAG.
4145	DILX=JMS	XDILX	/CLEAR DONE FLAG, LOAD X, WAIT FOR SETTLE. /SET DONE, DO NOT CLEAR AC.
4151	DILY=JMS	XDILY	/CLEAR DONE FLAG, LOAD Y, WAIT FOR SETTLE. /SET DONE, DO NOT CLEAR AC.
4155	DIXY=JMS	XDIXY	/CLEAR DONE FLAG, INTENSIFY, SET DONE.
4161	DILE=JMS	XDILE	/LOAD ENABLE REGISTER, CLEAR AC.
4165	DIRE=JMS	XDIRE	/TRANSFER ENABLE TO AC.
4025	SETUP=JMS	PRESET	
4101	ERROR=JMS	FAIL	
6007	CAF=6007		
7402	XX=7402		

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```

0001      0001      *1
0001 5402      JMP I   RETURN
0002 0000      RETURN, 0
0003 0000      SUM1, 0
0004 0000      ERSWIT, 0
0005 0000      TALLY, 0

      0020      *20
0020 0213      DISRET, DISMSG           /RETURN TO DISPATCH ROUTINE

0021 0000      TALLYA, 0
0022 0000      NXTST, 0
0023 0000      GETBAK, 0
0024 0000      DELAY, 0

      //HOUSEKEEPING ROUTINE FOR CONTROL LOGIC TEST//
0025 0000      PRESET, 0
0026 7200      CLA
0027 3004      DCA      ERSWIT
0030 1177      TAD      (5000
0031 3005      DCA      TALLY
0032 4036      JMS      CKSW7
0033 2035      ISZ      MSGPNT
0034 5425      JMP I    PRESET

      //ERROR MESSAGE POINTER//
0035 0062      MSGPNT, ERRMSG

      //ROUTINE TO CHECK LOOP BIT, S,R,7//
0036 0000      CKSW7, 0           /GET S,R,
0037 7604      LAS           /GET S,R,
0040 0176      AND      (20      /MASK BIT 7
0041 7650      SNA CLA      /S,R,7=0?
0042 5420      JMP I    DISRET      /YES, RETURN TO DISPATCH
0043 5436      JMP I    CKSW7      /NO, LOOP IN CURRENT TEST

```

```
//ROUTINE TO SELECT CHANNEL FOR VR1477
0044 0000 SELCHN. 0
0045 7604 LAS /GET S,R
0046 0175 AND (0400 /MASK BIT 3
0047 7640 SZA CLA /SW 3=
0050 5053 JMP SEL1
0051 4161 DILE /0, SELECT CHANNEL 1
0052 5444 JMP I SELCHN
0053 1174 SEL1. TAD (0002
0054 4161 DILE /1, SELECT CHANNEL 2
0055 5444 JMP I SELCHN

//SUBROUTINE CONTAINING IOT TO BE EXECUTED//
//
0056 0000 IOTT. 0
0057 7000 7000 /MODIFIED TO CONTAIN IOT
0060 7000 7000
0061 5456 JMP I IOTT
```

//ERROR MESSAGE LINKS//

0062	3676	ERRMSG,	EMSG1
0063	3717		EMSG1A
0064	3747		EMSG1B
0065	4001		EMSG1C
0066	4031		EMSG2
0067	4060		EMSG3
0070	4110		EMSG4
0071	4140		EMSG5
0072	4161		EMSG6
0073	4205		EMSG7
0074	4240		EMSG8
0075	4266		EMSG9
0076	4313		EMSG10
0077	4346		EMSG11
0100	4370		EMSG12

//SUBROUTINE TO HANDLE ERROR-SCOPE LOOPING//

0101	0000	FAIL,	0
0102	7200		CLA
0103	1004	TAD	ERSWIT
0104	7650	SNA	CLA
0105	4122	JMS	ERTYPE
0106	1173	TAD	(7777
0107	3004	DCA	ERSWIT
0110	1501	TAD I	FAIL
0111	3023	DCA	GETBAK
0112	2101	ISZ	FAIL
0113	1501	TAD I	FAIL
0114	3022	DCA	NXTST
0115	7604	LAS	
0116	0172	AND	(0200
0117	7650	SNA	CLA
0120	5423	JMP I	GETBAK
0121	5422	JMP I	NXTST

//ERROR TYPEOUT SUBROUTINE//

0122	0000	ERTYPE,	0
0123	1435	TAD I	MSGPNT
0124	3126	DCA	EOUT
0125	4771	JMS	MESSAGE
0126	7402	HLT	
0127	5522	JMP I	ERTYPE

```

//IOT SUBROUTINES//
//
0130 0000 XDICL, 0
0131 6050 RDICL, 6050 /CLEAR ENABLES, FLAGS, DELAYS,
0132 5530 JMP I XDICL
0133 7402 HLT

0134 0000 XDICD, 0
0135 6051 RDICD, 6051 /CLEAR DONE FLAG,
0136 5534 JMP I XDICD
0137 7402 HLT

0140 0000 XDISD, 0
0141 6052 RDISD, 6052 /SKIP ON DONE FLAG,
0142 7410 SKP
0143 2140 ISZ XDISD
0144 5540 JMP I XDISD

0145 0000 XDILX, 0
0146 6053 RDILX, 6053 /LOAD X BUFFER,
0147 5545 JMP I XDILX
0150 7402 HLT

0151 0000 XDILY, 0
0152 6054 RDILY, 6054 /LOAD Y BUFFER,
0153 5551 JMP I XDILY
0154 7402 HLT

0155 0000 XDIXY, 0
0156 6055 RDIXY, 6055 /INTENSIFY,
0157 5555 JMP I XDIXY
0160 7402 HLT

0161 0000 XDILE, 0
0162 6056 RDILE, 6056 /LOAD ENABLE REGISTER,
0163 5561 JMP I XDILE
0164 7402 HLT

0165 0000 XDIRE, 0
0166 6057 RDIRE, 6057 /TRANSFER ENABLE TO AC,
0167 5565 JMP I XDIRE
0170 7402 HLT
```

```

0200      0200      *200
0200      6007      START, CAF
0201      2255      ISZ      TITLE
0202      5213      JMP      DISMSG
0203      4777      JMS      MESSAGE
0204      3200      MSG1
0205      4777      JMS      MESSAGE
0206      3224      MSG2
0207      4777      JMS      MESSAGE
0210      3236      MSG3
0211      4777      JMS      MESSAGE
0212      3266      MSG4

```

```

0213      6007      DISMSG, CAF
0214      4777      JMS      MESSAGE
0215      3322      MSG5

```

//MASTER DISPATCH ROUTINE FOR TEST SELECTION//

```

0216      7300      DISPATCH, CLA CLL
0217      7604      LAS
0220      0376      AND      (0040      /GET SWITCHES,
0221      7104      CLL RAL      /MASK SW6
0222      3321      DCA      HIOT      /SETUP IOT MODIFIER,
0223      4256      JMS      SETIOT      /GOTO IOT MODIFY ROUTINE,
0224      7604      LAS      /GET SWITCHES
0225      0375      AND      (20      /MASK TO CHECK S.R.7
0226      7450      SNA      /SH7=0?
0227      5216      JMP      DISPATCH      /YES, LOOP
0230      7604      LAS      /NO, PICK UP TEST NUMBER
0231      0374      AND      (17      /MASK FOR S.R.8=11
0232      1373      TAD      (JMP I TEST+1      /SETUP TEST SELECTED,
0233      3234      DCA      TEST

```

```

0234      0000      TEST,      0
0235      0213      DISMSG      /S,R.=0, NO TEST
0236      0600      CLTST      /S,R.=1, CONTROL LOGIC TEST
0237      0460      RMPTST      /S,R.=2, RAMP TEST
0240      0400      CALTST      /S,R.=3, DC CALIBRATION TEST
0241      2242      DISTST      /S,R.=4, DISPLAYED CALIBRATION TEST
0242      2400      CROTST      /S,R.=5, CROSSING DIAGONALS TEST
0243      1400      HORTST      /S,R.=6, HORIZONTAL FLYBACK TEST
0244      1434      VERTST      /S,R.=7, VERTICAL FLYBACK TEST
0245      1600      CORTST      /S,R.=10, CORNERS TEST
0246      2000      DBTST      /S,R.=11, DIAGONAL LINE TEST
0247      2023      VRBTST      /S,R.=12, VERTICAL BAR TEST
0250      2057      HRBTST      /S,R.=13, HORIZONTAL BAR TEST
0251      2200      PNTST      /S,R.=14, SINGLE POINT PLOT TEST
0252      0213      DISMSG      /S,R.=15, NO TEST
0253      0213      DISMSG      /S,R.=16, NO TEST
0254      0213      DISMSG      /S,R.=17, NO TEST

```

```

0255      7777      TITLE,      7777      /TYPE OUT HEADER ONE TIME ONLY,

```

//ROUTINE TO MODIFY ALL IOT'S//

//

0256	0000	SETIOT, 0	
0257	1131	TAD	RDICL
0260	0320	AND	K7077
0261	1321	TAD	MIOT
0262	3131	DCA	RDICL
0263	1135	TAD	RDICD
0264	0320	AND	K7077
0265	1321	TAD	MIOT
0266	3135	DCA	RDICD
0267	1141	TAD	RDISD
0270	0320	AND	K7077
0271	1321	TAD	MIOT
0272	3141	DCA	RDISD
0273	1146	TAD	RDILX
0274	0320	AND	K7077
0275	1321	TAD	MIOT
0276	3146	DCA	RDILX
0277	1152	TAD	RDILY
0300	0320	AND	K7077
0301	1321	TAD	MIOT
0302	3152	DCA	RDILY
0303	1156	TAD	RDIXY
0304	0320	AND	K7077
0305	1321	TAD	MIOT
0306	3156	DCA	RDIXY
0307	1162	TAD	RDILE
0310	0320	AND	K7077
0311	1321	TAD	MIOT
0312	3162	DCA	RDILE
0313	1166	TAD	RDIRE
0314	0320	AND	K7077
0315	1321	TAD	MIOT
0316	3166	DCA	RDIRE
0317	5656	JMP I	SETIOT
0320	7077	K7077,	7077
0321	0000	MIOT,	0

0373 5635
 0374 0017
 0375 0020
 0376 0040
 0377 4600
 0400

PAGE

//DC CALIBRATION ROUTINE//

//

0400	6007	CALTST,	CAF		
0401	4777	JMS	MSG6	MESSAGE	
0402	3332				
0403	7200	CALIB,	CLA		
0404	1376		TAD	(=21	/INITIALIZE
0405	3256		DCA	CALCNT	/COUNTER
0406	7604		LAS		/GET S.R.
0407	0375		AND	(2000	/CHECK SW1
0410	7650		SNA	CLA	/TO DETERMINE X OR Y AXIS
0411	5215		JMP	CALX	/GOTO X AXIS SETUP
0412	1374	CALY,	TAD	(DILY	/SETUP Y AXIS
0413	3057		DCA	IOTT+1	/SAVE Y IOT
0414	5217		JMP	GETVAL	/GOTO BEGIN OF CAL ROUTINE
0415	1373	CALX,	TAD	(DILX	/SETUP X AXIS
0416	3057		DCA	IOTT+1	/SAVE X IOT
0417	1372	GETVAL,	TAD	(TABLE	/INITIALIZE POINTER
0420	3257		DCA	PNTR1	/FOR CALIBRATION PICKUP
0421	1657	UPDVAL,	TAD	PNTR1	/PICKUP CAL VALUE
0422	3003		DCA	SUM1	/SAVE IT
0423	4771		JMS	CALSND	/OUTPUT VALUE TO TTY
0424	1003		TAD	SUM1	/GET CAL VALUE AGAIN.
0425	4056		JMS	IOTT	/SEND TO DAC
0426	7402		HLT		/HALT AT PRESENT CAL VALUE.
0427	4036		JMS	CKSW7	/CHECK TEST LOOP SWITCH.
0430	2257		ISE	PNTR1	/INCREMENT POINTER
0431	2256		ISE	CALCNT	/HAS PRESENT AXIS CALLED OUT?
0432	5221		JMP	UPDVAL	/NO, GET NEXT VALUE
0433	5203		JMP	CALIB	/YES, RETURN TO BEGIN OF ROUTINE.

```
                //CALIBRATION TABLE//
0434 0777      TABLE, 0777
0435 0776      0776
0436 0775      0775
0437 0773      0773
0440 0767      0767
0441 0757      0757
0442 0740      0740
0443 0737      0737
0444 0700      0700
0445 0677      0677
0446 0600      0600
0447 0577      0577
0450 0400      0400
0451 0377      0377
0452 0000      0000
0453 0777      0777
0454 1000      1000

0455 0000      SWITCH, 0
0456 0000      CALCNT, 0
0457 0000      PNTRI, 0
```

```

//RAMP TEST//
//
0460 0007 RMPTST, CAF
0461 4777 JMS MESSAGE
0462 3346 MSG7
0463 7300 RAMP, CLA CLL
0464 7604 LAS /PICK UP S'R'
0465 0375 AND (2000 /CHECK SW1, TO DETERMINE
0466 7650 SNA CLA /X OR Y AXIS
0467 5273 JMP XRAMP /GOTO X AXIS SETUP
0470 1374 TAD (DILY /SETUP Y AXIS
0471 3057 DCA IOTT+1 /IOT
0472 5275 JMP RAMPA-1 /GOTO RAMP ROUTINE
0473 1373 XRAMP, TAD (DILX /SETUP X AXIS
0474 3057 DCA IOTT+1 /IOT
0475 1370 TAD (1000 /RAMP ORIGIN
0476 4056 RAMPA, JMS IOTT /SETUP DAC SELECTED
0477 1367 TAD (1 /INCREMENT RAMP
0500 3304 DCA SAVIT
0501 4036 JMS CKSW7 /CHECK TEST LOOP SWITCH
0502 1304 TAD SAVIT
0503 5276 JMP RAMPA /CONTINUE RAMP

0504 0000 SAVIT, 0

0567 0001
0570 1000
0571 3000
0572 0434
0573 4145
0574 4151
0575 2000
0576 7757
0577 4600
0600

```

```

//CONTROL LOGIC TEST//
//
0600 6007 CLTST, CAF
0601 4777 JMS MESSAGE
0602 3355 MSG8
0603 7200 CLA
0604 1376 TAD (-200
0605 3021 DCA TALLYA
/
/CHECK DATA TRANSFERS, AC AND ENABLE REGISTERS.
/
0606 4025 SETUP
0607 1375 TAD (ERRMSG
0610 3035 DCA MSGPNT
0611 7300 CLT1, CLA CLL
0612 1374 TAD (7777 /BRING AC TO ALL ONES,
0613 4161 DILE /TRANS, TO ENABLE AND CLR AC.
0614 7650 SNA CLA
0615 5221 JMP ,+4 /AC CLEAR, CONTINUE.
0616 4101 ERROR /DILE FAILED TO CLR AC.
0617 0611 CLT1 /ERROR-SCOPE LOOP ADDRESS.
0620 0623 CLT1A-1 /NEXT TEST.
0621 2005 ISZ TALLY /TEST LOOP COUNT.
0622 5211 JMP CLT1 /RETURN.
/
0623 4025 SETUP
0624 7300 CLT1A, CLA CLL
0625 1373 TAD (0002
0626 4161 DILE /SET CHANNEL P/F = 1.
0627 4165 DIRE /READ ENABLE INTO AC.
0630 0373 AND (0002 /MASK TO CHECK FOR CHANNEL = 1.
0631 7640 SZA CLA
0632 5236 JMP ,+4 /CHANNEL SET AND READ BACK.
0633 4101 ERROR /FAILED, CHNL NOT SET OR NOT READ BACK.
0634 0624 CLT1A /ERROR-SCOPE LOOP ADDRESS.
0635 0640 CLT1B-1 /NEXT TEST.
0636 2005 ISZ TALLY /TEST LOOP COUNT.
0637 5224 JMP CLT1A /RETURN.
/

```

```

0640 4025      SETUP
0641 7300      CLT1B, CLA CLL
0642 1372      TAD      (0001
0643 4161      DILE
0644 4165      DIRE
0645 0372      AND      (0001
0646 7640      SZA CLA
0647 5253      JMP      ,+4
0650 4101      ERROR
0651 0641      CLT1B
0652 0655      CLT1C-1
0653 2005      ISZ      TALLY
0654 5241      JMP      CLT1B
/
0655 4025      SETUP
0656 7300      CLT1C, CLA CLL
0657 1373      TAD      (0002
0660 4161      DILE
0661 4165      DIRE
0662 0373      AND      (0002
0663 7640      SZA CLA
0664 5270      JMP      ,+4
0665 4101      ERROR
0666 0656      CLT1C
0667 0672      CLT2-1
0670 2005      ISZ      TALLY
0671 5256      JMP      CLT1C
/
/ CHECK THAT DICL WILL CLEAR ENABLES
/
0672 4025      SETUP
0673 7300      CLT2, CLA CLL
0674 1371      TAD      (4003
0675 4161      DILE
0676 4130      DICL
0677 4165      DIRE
0700 7650      SNA CLA
0701 5305      JMP      ,+4
0702 4101      ERROR
0703 0673      CLT2
0704 0707      CLT3-1
0705 2005      ISZ      TALLY
0706 5273      JMP      CLT2
/SET INTERRUPT ENABLE.
/READ ENABLE INTO AC.
/MASK TO CHECK FOR INT. ENABLE.
/INT. SET AND READ BACK.
/FAILED, INT. NOT SET OR READ BACK.
/ERROR-SCOPE LOOP ADDRESS.
/NEXT TEST.
/TEST LOOP COUNT.
/RETURN.
/SET CHANNEL.
/READ ENABLE INTO AC.
/MASK TO CHECK FOR CHANNEL BIT.
/CHANNEL SET AND READ BACK.
/FAILED, CHNL NOT SET OR READ BACK.
/ERROR-SCOPE LOOP ADDRESS.
/NEXT TEST.
/TEST LOOP COUNT.
/RETURN.
/SET ENABLES
/CLEAR ENABLES
/READ ENABLE STATUS INTO AC.
/ENABLES CLEARED. CONTINUE.
/FAILED, ENABLES NOT CLEARED.
/ERROR-SCOPE LOOP ADDRESS.
/NEXT TEST.
/TEST LOOP COUNT.
/RETURN.

```

```

/
/CHECK THAT DILX WILL SET DONE AND NOT CLEAR AC.
/
0737 4025          SETUP
0710 7300          CLT3. CLA CLL
0711 1374          TAD      (7777
0712 3024          DCA      DELAY
0713 7604          LAS
0714 0370          AND      (1000
0715 7640          SZA CLA
0716 5321          JMP      ,+3
0717 1367          TAD      (-16
0720 3024          DCA      DELAY
0721 1374          TAD      (7777
0722 4134          DICO
0723 4145          DILX
0724 2024          ISZ      DELAY
0725 5324          JMP      ,+1
0726 4140          DISD
0727 7410          SKP
0730 5334          JMP      ,+4
0731 4101          ERROR
0732 0710          CLT3
0733 1000          CLT4-1
0734 7440          SZA
0735 5341          JMP      ,+4
0736 4101          ERROR
0737 0710          CLT3
0740 1000          CLT4-1
0741 2005          ISZ      TALLY
0742 5310          JMP      CLT3
0743 5770          JMP      CLT4-1

/SETUP VALUE OF
/DELAY FOR VR03A SCOPE.
/GET S,R.
/CHECK BIT 2 FOR SCOPE SELECTION.
/SW 2 = 0, SETUP FOR VR14.
/SW 2 = 1, SETUP FOR VR03A.
/SETUP VALUE OF
/DELAY FOR VR14.
/ALL 1'S TO AC.
/CLEAR DONE.
/LOAD X BUFFER.
/WAIT.
/SKIP ON DONE.
/DONE SET, CONTINUE.
/FAILED, DONE WAS NOT SET.
/ERROR-SCOPE LOOP ADDRESS.
/NEXT TEST.
/WAS AC CLEARED?
/NO, CONTINUE.
/YES, FAILED.
/ERROR-SCOPE LOOP ADDRESS.
/NEXT TEST.
/TEST LOOP COUNT.
/RETURN.
/NEXT TEST.

0767 7762
0770 1000
0771 4003
0772 0001
0773 0002
0774 7777
0775 0062
0776 7600
0777 4600
1000

```

```

/
/CHECK THAT DILY WILL SET DONE AND NOT CLEAR AC.
/
1000 4025          SETUP
1001 7300          CLT4, CLA CLL
1002 1377          TAD      (7777          /SETUP VALUE OF
1003 3024          DCA      DELAY        /DELAY FOR VR03A SCOPE.
1004 7604          LAS      /GET S,R.
1005 0376          AND      (1000        /CHECK BIT 2 FOR SCOPE SELECTION.
1006 7640          SZA CLA          /S,R, 2=0, SETUP FOR VR14.
1007 5212          JMP      ,+3          /S,R, 2=1, SETUP FOR VR03A.
1010 1375          TAD      (=16        /SETUP VALUE OF
1011 3024          DCA      DELAY        /DELAY FOR VR14.
1012 1377          TAD      (7777        /ALL 1'S TO AC.
1013 4134          DICO          /CLEAR DONE.
1014 4151          DILY          /LOAD Y BUFFER.
1015 2024          ISZ      DELAY
1016 5215          JMP      ,=1
1017 4140          DISD          /SKIP ON DONE.
1020 7410          SKP
1021 5225          JMP      ,+4          /DONE SET, CONTINUE.
1022 4101          ERROR        /FAILED, DONE WAS NOT SET.
1023 1001          CLT4          /ERROR=SCOPE LOOP ADDRESS.
1024 1034          CLT5=1        /NEXT TEST.
1025 7440          SZA          /HAS AC CLEARED?
1026 5232          JMP      ,+4          /NO, CONTINUE.
1027 4101          ERROR        /YES, FAILED.
1030 1001          CLT4          /ERROR=SCOPE LOOP ADDRESS.
1031 1034          CLT5=1        /NEXT TEST.
1032 2005          ISZ      TALLY      /TEST LOOP COUNT.
1033 5201          JMP      CLT4      /RETURN.

```



```

/
/CHECK THAT DIXY WILL SET DONE,
/
1034 4025          SETUP
1035 7300          CLT5, CLA CLL
1036 4130          DIDL          /CLEAR FLAGS,
1037 4155          DIXY          /INTENSIFY AND SET DONE,
1040 4140          DISD          /SKP ON DONE,
1041 7410          SKP
1042 5246          JMP          ,+4          /DONE SET, CONTINUE,
1043 4101          ERROR          /FAILED, DONE NOT SET,
1044 1035          CLT5          /ERROR-SCOPE LOOP ADDRESS,
1045 1050          CLT6-1        /NEXT TEST,
1046 2005          ISZ          TALLY          /TEST LOOP COUNT,
1047 5235          JMP          CLT5          /RETURN,
/
/CHECK THAT DICD CLEARS DONE FLAG,
/
1050 4025          SETUP
1051 7300          CLT6, CLA CLL
1052 4155          DIXY          /SET DONE,
1053 4134          DICD          /CLEAR DONE,
1054 4165          DIRE          /READ ENABLE STATUS INTO AC,
1055 7650          SNA CLA
1056 5262          JMP          ,+4          /DONE CLEARED, CONTINUE,
1057 4101          ERROR          /FAILED, DONE NOT CLEARED,
1060 1051          CLT6          /ERROR-SCOPE LOOP ADDRESS,
1061 1064          CLT7-1        /NEXT TEST,
1062 2005          ISZ          TALLY          /TEST LOOP COUNT,
1063 5251          JMP          CLT6          /RETURN,
```

```

/
/CHECK THAT DISD WILL SKIP ON DONE FLAG AND NOT CLEAR FLAG.
/
1064 4025          SETUP
1065 7300  CLT7,   CLA CLL
1066          DIXY          /SET DONE.
1067 4140          DISD      /SKP ON DONE.
1070 7610          SKP CLA
1071 5275          JMP      ,+4      /SKIP WORKED, CONTINUE.
1072 4101          ERROR      /FAILED, DISD DID NOT CAUSE SKP ON DONE.
1073 1065          CLT7        /ERROR=SCOPE LOOP ADDRESS.
1074 1113          CLT8=1      /NEXT TEST.
1075 4165          DIRE      /READ ENABLE STATUS INTO AC.
1076 7640          SZA CLA      /DONE = 1?
1077 5303          JMP      ,+4      /YES, CONTINUE.
1100 4101          ERROR      /NO, FAILED.
1101 1065          CLT7        /ERROR=SCOPE LOOP ADDRESS.
1102 1113          CLT8=1      /NEXT TEST.
1103 4134          DICO      /CLEAR DONE.
1104 4140          DISD      /SKP ON DONE.
1105 5311          JMP      ,+4
1106 4101          ERROR      /FAILED, DISD SKIPPED ON DONE=0.
1107 1065          CLT7        /ERROR=SCOPE LOOPE ADDRESS.
1110 1113          CLT8=1      /NEXT TEST.
1111 2005          ISZ      TALLY  /TEST LOOP COUNT.
1112 5265          JMP      CLT7  /RETURN.
/
/CHECK THAT INTERRUPT ENABLE REGISTER ALONE WILL
/NOT CAUSE AN INTERRUPT.
1113 4025          SETUP
1114 7300  CLT8,   CLA CLL
1115 1374          TAD      (ERR8   /GET RETURN ADDRESS.
1116 3002          DCA      RETURN  /SETUP RETURN ADDRESS.
1117 6007          CAF      /CLEAR ALL.
1120 1373          TAD      (0001   /SETUP AC TO
1121 4161          DILE      /ENABLE INTERRUPT REGISTER.
1122 6001          ION      /TURN INTERRUPT ON.
1123 7000          NOP      /WAIT.
1124 6002          IOF      /TURN INTERRUPT OFF.
1125 5331          JMP      ,+4      /NO INTERRUPT OCCURRED, CONTINUE.
1126 4101          ERR8,   ERROR      /FAILED, INTERRUPT TOOK PLACE.
1127 1114          CLT8        /ERROR=SCOPE LOOP ADDRESS.
1130 1200          CLT9=1      /NEXT TEST.
1131 2005          ISZ      TALLY  /TEST LOOP COUNT.
1132 5314          JMP      CLT8   /RETURN.
1133 5772          JMP      CLT9=1 /NEXT TEST.

1172 1200
1173 0001
1174 1126
1175 7762
1176 1000
1177 7777
1200

```

```

/
/CHECK THAT DONE REGISTER ALONE WILL NOT CAUSE
/AN INTERRUPT.
1200 4025          SETUP
1201 7300  CLT9,  CLA CLL
1202 1377          TAD      (ERR9      /GET RETURN ADDRESS.
1203 3002          DCA      RETURN    /SETUP RETURN ADDRESS.
1204 6007          CAF              /CLEAR ALL.
1205 1376          TAD      (4000     /SETUP AC TO
1206 4161          DILE              /ENABLE DONE FLAG.
1207 6001          ION              /TURN INTERRUPT ON.
1210 7000          NOP              /WAIT.
1211 6002          IOF              /TURN INTERRUPT OFF.
1212 5216          JMP      ,+4      /NO INTERRUPT OCCURRED, CONTINUE.
1213 4101  ERR9,  ERROR              /FAILED, INTERRUPT TOOK PLACE.
1214 1201          CLT9              /ERROR-SCOPE LOOP ADDRESS.
1215 1220          CLT10-1          /NEXT TEST.
1216 2005          ISZ      TALLY    /TEST LOOP COUNT.
1217 5201          JMP      CLT9     /RETURN.
/
/CHECK THAT DONE AND INTERRUPT ENABLE WILL CAUSE AN
/INTERRUPT.
1220 4025          SETUP
1221 7300  CLT10,  CLA CLL
1222 1375          TAD      (OK10     /GET RETURN ADDRESS.
1223 3002          DCA      RETURN    /SETUP RETURN ADDRESS.
1224 1374          TAD      (0001     /SETUP AC TO
1225 4155          DIXY              /ENABLE DONE
1226 4161          DILE              /AND INTERRUPT.
1227 6001          ION              /TURN INTERRUPT ON.
1230 7000          NOP              /WAIT.
1231 6002          IOF              /TURN INTERRUPT OFF.
1232 4101          ERROR              /FAILED, INTERRUPT DID NOT TAKE PLACE.
1233 1221          CLT10              /ERROR-SCOPE LOOP ADDRESS.
1234 1237          CLT11-1          /NEXT TEST.
1235 2005  OK10,  ISZ      TALLY    /TEST LOOP COUNTER.
1236 5221          JMP      CLT10    /RETURN.

```

```

/
/CHECK THAT DILX WILL CLEAR DONE'
/
1237 4025          SETUP
1240 7300      CLT11, CLA CLL
1241 1146          TAD      RDILX
1242 3250          DCA      CLT11A
1243 1166          TAD      RDIRE
1244 3251          DCA      CLT11A+1
1245 4145          DILX
1246 4140          DISD
1247 5246          JMP      ,=1
1250 7402      CLT11A, XX
1251 7402          XX
1252 0376          AND      (4000
1253 7450          SNA
1254 5260          JMP      ,+4
1255 4101          ERROR
1256 1240          CLT11
1257 1262          CLT12=-1
1260 2005          ISZ      TALLY
1261 5240          JMP      CLT11

```

```

/
/CHECK THAT DILY WILL CLEAR DONE'
/
1262 4025          SETUP
1263 7300      CLT12, CLA CLL
1264 1152          TAD      RDILY
1265 3273          DCA      CLT12A
1266 1166          TAD      RDIRE
1267 3274          DCA      CLT12A+1
1270 4151          DILY
1271 4140          DISD
1272 5271          JMP      ,=1
1273 7402      CLT12A, XX
1274 7402          XX
1275 0376          AND      (4000
1276 7450          SNA
1277 5303          JMP      ,+4
1300 4101          ERROR
1301 1263          CLT12
1302 0606          CLT1=-3
1303 2005          ISZ      TALLY
1304 5263          JMP      CLT12
1305 2021          ISZ      TALLYA
1306 5773          JMP      CLT1=-3
1307 5772          JMP      CLTST

```

```

1372 0600
1373 0606
1374 0001
1375 1235
1376 4000
1377 1213
1400

```

//HORIZONTAL LINE SEGMENT TEST
 //DISPLAY FOUR HORIZONTAL LINE SEGMENTS
 //CHECK FOR FLYBACK TRACE

1400	6007	HORTST,	CAF		
1401	4777	JMS	MESSAGE		
1402	3404	MSG10			
1403	7300	CLA	CLL		
1404	4776	JMS	SETHOR		/SETUP FOR HORIZONTAL LINES.
1405	4044	HORFLY,	JMS	SELCHN	/CHECK FOR VR=14 CHANNEL.
1406	7200	CLA			/DISPLAY HORIZONTAL LINE AT Y=1001 (RIGHT).
1407	1375	TAD	(1001		/NON-VARIABLE AXIS ORIGIN (Y).
1410	4774	JMS	PLINE		/ROUTINE TO INCREMENT X AXIS.
1411	7377	-401			/PLOT COUNTER.
1412	1001	1001			/ORIGIN OF X AXIS.
1413	7200	CLA			/DISPLAY HORIZONTAL LINE AT Y=777 (RIGHT).
1414	1373	TAD	(0777		/NON-VARIABLE AXIS ORIGIN (Y).
1415	4774	JMS	PLINE		/ROUTINE TO INCREMENT X AXIS.
1416	7377	-401			/PLOT COUNTER.
1417	1001	1001			/ORIGIN OF X AXIS.
1420	7200	CLA			/DISPLAY HORIZONTAL LINE AT Y=1001 (LEFT).
1421	1375	TAD	(1001		/NON-VARIABLE AXIS ORIGIN (Y).
1422	4772	JMS	MLINE		/ROUTINE TO DECREMENT X AXIS.
1423	7377	-401			/PLOT COUNTER.
1424	0777	0777			/ORIGIN OF X AXIS.
1425	7200	CLA			/DISPLAY HORIZONTAL LINE AT Y=777 (LEFT).
1426	1373	TAD	(0777		/NON-VARIABLE AXIS ORIGIN (Y).
1427	4772	JMS	MLINE		/ROUTINE TO DECREMENT X AXIS.
1430	7377	-401			/PLOT COUNTER.
1431	0777	0777			/ORIGIN OF X AXIS.
1432	4036	JMS	CKSW7		/SW7=0, RETURN TO DISPATCH.
1433	5205	JMP	HORFLY		/SW7=1, CONTINUE IN CURRENT TEST.

```

//VERTICAL LINE SEGMENT TEST
//DISPLAY FOUR VERTICAL LINE SEGMENTS
//CHECK FOR FLYBACK TRACE

1434 6007 VERTST, CAF
1435 4777 JMS MESSAGE
1436 3371 MSG9
1437 7300 CLA CLL
1440 4771 JMS SETVER /SETUP FOR VERTICAL LINES;
1441 4044 VERFLY, JMS SELCHN /CHECK FOR VR=14 CHANNEL;
1442 7300 CLA CLL /DISPLAY VERTICAL LINE AT X=1001 (UP);
1443 1375 TAD (1001 /NON-VARIABLE AXIS ORIGIN (X);
1444 4774 JMS PLINE /ROUTINE TO INCREMENT Y AXIS;
1445 7377 -401 /PLOT COUNTER;
1446 1001 1001 /ORIGIN OF Y AXIS;
1447 7200 CLA /DISPLAY VERTICAL LINE AT X=0777 (UP);
1450 1373 TAD (0777 /NON-VARIABLE AXIS ORIGIN (X);
1451 4774 JMS PLINE /ROUTINE TO INCREMENT Y AXIS;
1452 7377 -401 /PLOT COUNTER;
1453 1001 1001 /ORIGIN OF Y AXIS;
1454 7200 CLA /DISPLAY VERTICAL LINE AT X=1001 (DOWN);
1455 1375 TAD (1001 /NON-VARIABLE AXIS ORIGIN (X);
1456 4772 JMS MLINE /ROUTINE TO DECREMENT Y AXIS;
1457 7377 -401 /PLOT COUNTER;
1460 0777 0777 /ORIGIN OF Y AXIS;
1461 7200 CLA /DISPLAY VERTICAL LINE AT X=0777 (DOWN);
1462 1373 TAD (0777 /NON-VARIABLE AXIS ORIGIN (X);
1463 4772 JMS MLINE /ROUTINE TO DECREMENT Y AXIS;
1464 7377 -401 /PLOT COUNTER;
1465 0777 0777 /ORIGIN OF Y AXIS;
1466 4036 JMS CKSW7 /SW7=0, RETURN TO DISPATCH;
1467 5241 JMP VERFLY /SW7=1, CONTINUE IN CURRENT TEST.

1571 2600
1572 2416
1573 0777
1574 2443
1575 1001
1576 2613
1577 4600
1600

```

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//CORNERS - ROUTINE TO DISPLAY FOUR CORNERS

//WITH INTERSECTING DIAGONAL LINE SEGMENTS

//

1600	6007	CORTST,	CAF		
1601	4777		JMS	MESSAGE	
1602	3420		MSG11		
1603	7300		CLA	CLL	
1604	4044	CORNER,	JMS	SELCHN	/CHECK FOR VR=14 CHANNEL
1605	4776		JMS	SETVER	/SETUP FOR VERTICAL LINES
1606	7300		CLA	CLL	
1607	1375		TAD	(1001	/X AXIS ORIGIN
1610	4774		JMS	PLINE	/PLOT A VERTICAL LINE AT X=1001(UP)
1611	7577		=201		/PLOT COUNTER
1612	1001		1001		/Y AXIS ORIGIN
1613	7200		CLA		
1614	1373		TAD	(0777	
1615	4772		JMS	MLINE	/PLOT A VERTICAL LINE AT X=777(DOWN)
1616	7577		=201		
1617	0777		0777		
1620	7200		CLA		
1621	1375		TAD	(1001	
1622	4772		JMS	MLINE	/PLOT A VERTICAL LINE AT X=1001(DOWN)
1623	7577		=201		
1624	0777		0777		
1625	7200		CLA		
1626	1373		TAD	(0777	
1627	4774		JMS	PLINE	/PLOT A VERTICAL LINE AT X=777(UP)
1630	7577		=201		
1631	1001		1001		
1632	4771		JMS	SETHOR	/SETUP FOR HORIZONTAL LINES
1633	7200		CLA		
1634	1373		TAD	(0777	/Y AXIS ORIGIN
1635	4774		JMS	PLINE	/PLOT HORIZONTAL LINE AT Y=777(RIGHT)
1636	7577		=201		
1637	1001		1001		
1640	7200		CLA		
1641	1375		TAD	(1001	
1642	4774		JMS	PLINE	/PLOT HORIZONTAL LINE AT Y=1001(RIGHT)
1643	7577		=201		
1644	1001		1001		
1645	7200		CLA		
1646	1373		TAD	(0777	
1647	4772		JMS	MLINE	/PLOT HORIZONTAL LINE AT Y=777(LEFT)
1650	7577		=201		
1651	0777		0777		

1652	7200	CLA		
1653	1375	YAD	(1001	
1654	4772	JMS	MLINE	/PLOT HORIZONTAL LINE AT Y=1001(LEFT)
1655	7577	=201		
1656	0777	0777		
1657	7200	CLA		
1660	4770	JMS	DIAG1	/PLOT DIAGNOL LINE (LOWER LEFT)
1661	7577	=201		
1662	1001	1001		
1663	7200	CLA		
1664	4770	JMS	DIAG1	/PLOT DIAGNOL LINE (UPPER RIGHT)
1665	7577	=201		
1666	0577	0577		
1667	7200	CLA		
1670	4767	JMS	DIAG2	/PLOT DIAGNOL LINE (UPPER LEFT)
1671	7577	=201		
1672	1001	1001		/X ORIGIN,
1673	0777	0777		/Y ORIGIN,
1674	7200	CLA		
1675	4767	JMS	DIAG2	/PLOT DIAGNOL LINE (LOWER RIGHT)
1676	7577	=201		
1677	0577	0577		
1700	1201	1201		
1701	4036	JMS	CRSW7	
1702	5204	JMP	CORNER	
1767	2660			
1770	2626			
1771	2613			
1772	2416			
1773	0777			
1774	2443			
1775	1001			
1776	2600			
1777	4600			
	2000			

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//ROUTINE TO DISPLAY DIAGONALS

//

2000	6007	DBTST,	CAF		
2001	4777	JMS	MESSAGE		
2002	3633	MSG19			
2003	7300	CLA CLL			
2004	4044	DIABIS,	JMS	SELCHN	/CHECK FOR VR=14 CHANNEL
2005	7604	LAS			/GET S,R.
2006	0376	AND	(0100		/MASK TO CHECK SW5
2007	7640	SZA CLA			/SW5 = 0, PLOT LL TO UR DIAGONAL
2010	5215	JMP	,+5		/SW5 = 1, PLOT UL TO LR DIAGONAL
2011	4775	JMS	DIAG1		/PLOT LL TO UR DIAGONAL
2012	5777	=2001			
2013	1001	1001			
2014	5221	JMP	,+5		
2015	4774	JMS	DIAG2		/PLOT UL TO LR DIAGONAL
2016	6000	=2000			
2017	1001	1001			
2020	0777	0777			
2021	4036	JMS	CKSW7		/SW7=0, RETURN TO DISPATCH
2022	5204	JMP	DIABIS		/SW7=1, CONTINUE PLOT

//ROUTINE TO MOVE A VERTICAL BAR HORIZONTALLY.

```

2023 6007 VRBTST, CAF //
2024 4777 JMS MESSAGE
2025 3431 MSG12
2026 7300 CLA CLL
2027 4044 JMS SELCHN /CHECK FOR VR-14 CHANNEL.
2030 4773 JMS SETVER /SETUP FOR VERTICAL LINES.
2031 7200 CLA
2032 1372 TAD (-2000
2033 3255 DCA HORCNT /SET UP X AXIS COUNTER.
2034 1371 TAD (1001 /SET UP X ORIGIN.
2035 3256 DCA XVERT
2036 7200 VERBAR, CLA
2037 1256 TAD XVERT /GER X COORDINATE.
2040 4770 JMS PLINE /PLOT VERTICAL BAR.
2041 5777 -2001 /COUNT.
2042 1001 1001 /Y COORDINATE.
2043 7604 LAS /GET S,R.
2044 0376 AND (0100 /MASK BIT 5
2045 7640 SZA CLA /S,R, 5 = 0 CONTINUE LINE MOVE.
2046 5236 JMP VERBAR /S,R, 5 = 1 HALT LINE MOVEMENT.
2047 2256 ISZ XVERT /UPDATE X COORDINATE.
2050 7000 NOP /UPDATE X AXIS COUNTER.

2051 2255 ISZ HORCNT /IS PLOT COMPLETE?
2052 5236 JMP VERBAR /NO. CONTINUE.
2053 4036 JMS CKSW7 /SW7=0, RETURN TO DISPATCH.
2054 5223 JMP VRBTST /SW7=1, CONTINUE IN CURRENT TEST.
2055 0000 HORCNT, 0
2056 0000 XVERT, 0

```

```

//ROUTINE TO MOVE A HORIZONTAL BAR VERTICALLY.
//
2057 6007   HRBST, CAF
2060 4777   JMS     MESSAGE
2061 3444   MSG13
2062 7300   CLA CLL
2063 4044   JMS     SELCHN   /CHECK FOR VR=14 CHANNEL.
2064 4767   JMS     SETHOR   /SETUP FOR HORIZONTAL LINES.
2065 7200   CLA
2066 1372   TAD     (-2000
2067 3311   DCA     VERCNT   /SETUP Y AXIS COUNTER.
2070 1371   TAD     (1001    /SETUP Y ORIGIN.
2071 3312   DCA     YVERT
2072 7200   HORBAR, CLA
2073 1312   TAD     YVERT    /GET Y COORDINATE.
2074 4770   JMS     PLINE    /PLOT HORIZONTAL BAR.
2075 5777   -2001          /COUNT.
2076 1001   1001          /X COORDINATE.
2077 7604   LAS          /GET S,R.
2100 0376   AND     (0100   /MASK BIT 5
2101 7640   SZA CLA      /SW 5 = 0 CONTINUE LINE MOVE.
2102 5272   JMP     HORBAR  /SW 5 = 1 HALT LINE MOVEMENT.
2103 2312   ISZ     YVERT   /UPDATE Y COORDINATE.
2104 7000   NOP          /UPDATE Y AXIS COUNTER.
2105 2311   ISZ     VERCNT  /IS PLOT COMPLETE?
2106 5272   JMP     HORBAR  /NO, CONTINUE.
2107 4036   JMS     CKSW7   /SW7=0, RETURN TO DISPATCH.
2110 5257   JMP     HRBST   /SW7=1, CONTINUE IN CURRENT TEST.
2111 0000   VERCNT, 0
2112 0000   YVERT, 0
2167 2613
2170 2443
2171 1001
2172 6000
2173 2600
2174 2660
2175 2626
2176 0100
2177 4600
2200

```

PAGE

```

//SINGLE POINT PLOT TEST
//ALL COORDINANTS FROM S,R.
//
2200 6007 PNTST, CAF
2201 4777 JMS MESSAGE
2202 3460 MSG14
2203 4777 JMS MESSAGE
2204 3476 MSG15
2205 7402 XX /HALT
2206 7604 LAS /GET X COORDINANT FROM SWITCHES.
2207 3240 DCA XPOINT /SAVE IT.
2210 4777 JMS MESSAGE
2211 3531 MSG16
2212 7402 XX /HALT
2213 7604 LAS /GET Y COORDINANT FROM SWITCHES.
2214 3241 DCA YPOINT /SAVE IT.
2215 4777 JMS MESSAGE
2216 3564 MSG17
2217 4777 JMS MESSAGE
2220 3602 MSG18
2221 7402 XX /HALT
2222 7200 SINPNT, CLA
2223 1240 TAD XPOINT /GET X COORDINANT.
2224 4145 DILX /LOAD X.
2225 4140 DISD /SKP ON DONE
2226 5225 JMP ,=1
2227 7200 CLA
2230 1241 TAD YPOINT /GET Y COORDINANT.
2231 4151 DILY /LOAD Y.
2232 4140 DISD /SKP ON DONE
2233 5232 JMP ,=1
2234 4155 DIXY /INTENSIFY
2235 4044 JMS SELCHN
2236 4036 JMS CKSW7
2237 5222 JMP SINPNT /SW7=0, RETURN TO DISPATCH.
/ SW7=1, CONTINUE IN CURRENT TEST.

2240 0000 XPOINT, 0
2241 0000 YPOINT, 0

```

//DISPLAYED CALIBRATION TEST

//

2242	6007	DISTST,	CAF		
2243	4777	JMS	MESSAGE		
2244	3656		MSG21		
2245	7340	DISCAL,	CLA CLL	CMA	/INITIALIZE
2246	3776		DCA	SWITCH	/SWITCH LOCATION.
2247	1375		TAD	(=13	/INITIALIZE
2250	3774		DCA	CALCNT	/COUNTER.
2251	7604		LAS		/GET S,R.
2252	0373		AND	(2000	/CHECK SW1
2253	7650		SNA CLA		/TO DETERMINE X OR Y AXIS.
2254	5257		JMP	,+3	
2255	4772		JMS	SETVER	/SW1=1, SETUP Y AXIS.
2256	7410		SKP		
2257	4771		JMS	SETHOR	/SW1=0, SETUP X AXIS.
2260	1370		TAD	(TABLEA	/INITIALIZE POINTER.
2261	3326		DCA	PNTR2	/FOR CALIBRATION PICKUP.
2262	1726		TAD I	PNTR2	/PICKUP CAL VALUE.
2263	3003		DCA	SUM1	
2264	4767		JMS	CALSDND	/OUTPUT LIMITS OF LINE TO TTY.
2265	1003	DISLOP,	TAD	SUM1	/GET BIT
2266	7040		CMA		/TO DETERMINE
2267	1366		TAD	(=1	/LENGTH OF LINE.
2270	3272		DCA	VARYCT	/SETUP PLOT COUNTER.
2271	4765		JMS	PLINE	/PLOT LINE.
2272	7402	VARYCT,	XX		/MODIFIED TO PLOT COUNT.
2273	0000		0000		/ORIGIN OF LINE.
2274	4036		JMS	CKSW7	/CHECK TEST LOOP SWITCH.
2275	4044		JMS	SELCHN	/CHECK VR-14 CHANNEL.
2276	7704		LAS CLL		/GET S,R.
2277	7006		RTL		/CHECK SW0
2300	0364		AND	(1	/TO SEE IF IT HAS
2301	1776		TAD	SWITCH	/CHANGED SINCE
2302	7640		SZA CLA		/LAST PASS.
2303	5265		JMP	DISLOP	/NO, CONTINUE IN PRESENT PLOT.
2304	1776		TAD	SWITCH	/YES, RESET
2305	7040		CMA		/LOCATION
2306	3776		DCA	SWITCH	/SWITCH.
2307	2326		ISZ	PNTR2	/INCREMENT POINTER.
2310	2774		ISZ	CALCNT	/HAS PRESENT AXIS CALLED OUT?
2311	5262		JMP	DISLOP+3	/NO, GET NEXT VALUE.
2312	5247		JMP	DISCAL+2	/YES, RETURN TO BEGIN OF ROUTINE.

2313	0000	TABLEA, 0000
2314	0001	0001
2315	0003	0003
2316	0007	0007
2317	0017	0017
2320	0037	0037
2321	0077	0077
2322	0177	0177
2323	0377	0377
2324	0777	0777
2325	1777	1777

2326	0000	PNTR2, 0
------	------	----------

2364	0001
2365	2443
2366	7777
2367	3000
2370	2313
2371	2613
2372	2600
2373	2000
2374	0456
2375	7765
2376	0455
2377	4600

2400	PAGE
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//CROSSING DIAGONALS TEST

//

2400	6007	CROTST,	CAF		
2401	4777	JMS	MESSAGE		
2402	3642	MSG20			
2403	7300	CLA CLL			
2404	4044	XCROS,	JMS	SELCHN	/CHECK FOR VR=14 CHANNEL.
2405	4776	JMS	DIAG1		/PLOT LL TO UR DIAGONAL.
2406	7000	-1000			/COUNT,
2407	1400	1400			/X AND Y ORIGINS.
2410	4775	JMS	DIAG2		/PLOT UL TO LR DIAGONAL.
2411	6777	-1001			/COUNT,
2412	1400	1400			/X ORIGIN,
2413	0400	0400			/Y ORIGIN,
2414	4036	JMS	CKSW7		/SW7=0, RETURN TO DISPATCH.
2415	5204	JMP	XCROS		/SW7=1, CONTINUE PLOT.

```
//SUBROUTINE TO DISPLAY A LINE (MINUS INCREMENTS)
//HORIZONTAL OR VERTICAL
//
```

2416	0000	MLINE,	0	
2417	7402	DISP3,	XX	/MODIFIED TO DISPLAY IOT,
2420	7200		CLA	
2421	1141		TAD	RDIS0 /SETUP
2422	3232		DCA	DISP4-3 /SKIP ON DONE IOT,
2423	1156		TAD	RDIY /SETUP
2424	3234		DCA	DISP4-1 /INTENSIFY IOT,
2425	1616		TAD I	MLINE /GET PLOT COUNT,
2426	3270		DCA	PLOTCT /SAVE IT,
2427	2216		ISZ	MLINE
2430	1616		TAD I	MLINE
2431	5235		JMP	DISP4
2432	6052		6052	/SKIP ON DONE,
2433	5232		JMP	, -1 /WAIT FOR DONE,
2434	6055		6055	/INTENSIFY
2435	7402	DISP4,	XX	/MODIFIED TO LOAD IOT,
2436	1374		TAD	(-1 /DECREMENT VARIABLE AXIS,
2437	2270		ISZ	PLOTCT /IS PLOT COMPLETE?
2440	5232		JMP	DISP4-3 /NO, CONTINUE PLOT,
2441	2216		ISZ	MLINE /YES, SETUP RETURN,
2442	5616		JMP I	MLINE /RETURN


```

                //SUBROUTINE TO DISPLAY A LINE (PLUS INCREMENTS)
                //HORIZONTAL OR VERTICAL
2443 0000 PLINE, 0
2444 7402 DISPI, XX /MODIFIED TO DISPLAY IOT.
2445 7200 CLA
2446 1141 TAD RDISD /SETUP
2447 3257 DCA DISP2=3 /SKIP ON DONE IOT.
2450 1156 TAD RDIY /SETUP
2451 3261 DCA DISP2=1 /INTENSIFY IOT.
2452 1643 TAD I PLINE /GET PLOT COUNT.
2453 3270 DCA PLOTCT /SAVE IT.
2454 2243 ISZ PLINE
2455 1643 TAD I PLINE /GET ORIGIN OF VARIABLE AXIS.
2456 5262 JMP DISP2
2457 6052 6052 /SKIP ON DONE.
2460 5257 JMP ,=1 /WAIT FOR DONE.
2461 6055 6055 /INTENSIFY
2462 7402 DISP2, XX /MODIFIED TO LOAD IOT.
2463 7001 IAC /INCREMENT VARIABLE AXIS.
2464 2270 ISZ PLOTCT /IS PLOT COMPLETE?
2465 5257 JMP DISP2=3 /NO, CONTINUE PLOT.
2466 2243 ISZ PLINE /YES, SETUP RETURN.
2467 5643 JMP I PLINE /RETURN

2470 0000 PLOTCT, 0
2574 7777
2575 2660
2576 2626
2577 4600
2600

```

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```

//SUBROUTINE TO SETUP DISPLAY IOT'S FOR VERTICAL PLOT//
2600 0000   SETVER, 0
2601 7200   CLA
2602 1146   TAD    RDILX
2603 3777   DCA    DISP1
2604 1152   TAD    RDILY
2605 3776   DCA    DISP2
2606 1146   TAD    RDILX
2607 3775   DCA    DISP3
2610 1152   TAD    RDILY
2611 3774   DCA    DISP4
2612 5600   JMP I   SETVER
  
```

```

//SUBROUTINE TO SETUP DISPLAY IOT'S FOR HORIZONTAL PLOT//
2613 0000   SETHOR, 0
2614 7200   CLA
2615 1152   TAD    RDILY
2616 3777   DCA    DISP1
2617 1146   TAD    RDILX
2620 3776   DCA    DISP2
2621 1152   TAD    RDILY
2622 3775   DCA    DISP3
2623 1146   TAD    RDILX
2624 3774   DCA    DISP4
2625 5613   JMP I   SETHOR
  
```

```

//SUBROUTINE TO DISPLAY A DIAGONAL LINE
//FROM LOWER LEFT TO UPPER RIGHT,
//
2626 0020      DIAG1, 0
2627 7300      CLA CLL
2630 1141      TAD      RDISD      /SETUP
2631 3245      DCA      DIAG1A=3    /SKIP ON DONE IOT,
2632 1156      TAD      RDIY      /SETUP
2633 3247      DCA      DIAG1A=1    /INTENSIFY IOT,
2634 1146      TAD      RDILX      /SETUP
2635 3250      DCA      DIAG1A      /LOAD X IOT,
2636 1152      TAD      RDILY      /SETUP
2637 3251      DCA      DIAG1A+1    /LOAD Y IOT,
2640 1626      TAD I      DIAG1      /SETUP COUNTER,
2641 3257      DCA      DIACNT
2642 2226      ISZ      DIAG1
2643 1626      TAD I      DIAG1
2644 5250      JMP      DIAG1A
2645 6052      6052      /SKIP ON DONE,
2646 5245      JMP      ,=1        /WAIT FOR DONE,
2647 6055      6055      /INTENSIFY
2650 6053      DIAG1A, 6053      /LOAD X
2651 6054      6054      /LOAD Y
2652 7001      IAC      /INCREMENT COORDINANT,
2653 2257      ISZ      DIACNT      /IS PLOT COMPLETE?
2654 5245      JMP      DIAG1A=3    /NO, CONTINUE PLOT,
2655 2226      ISZ      DIAG1      /YES, SETUP RETURN,
2656 5626      JMP I      DIAG1      /RETURN

2657 0000      DIACNT, 0

```

```
//SUBROUTINE TO DISPLAY A DIAGNOL LINE
//FROM UPPER LEFT TO LOWER RIGHT.
//
```

2660	0000	DIAG2,	0		
2661	7300		CLA CLL		
2662	1141		TAD	RDISD	/SETUP
2663	3303		DCA	DIAG2A=3	/SKIP ON DONE IOI.
2664	1156		TAD	RDIXY	/SETUP
2665	3305		DCA	DIAG2A=1	/INTENSIFY IOI.
2666	1146		TAD	RDILX	/SETUP
2667	3307		DCA	DIAG2A+1	/LOAD X IOI.
2670	1152		TAD	RDILY	/SETUP
2671	3312		DCA	DIAG2A+4	/LOAD Y IOI.
2672	1660		TAD I	DIAG2	/SETUP COUNTER.
2673	3257		DCA	DIACNT	
2674	2260		ISZ	DIAG2	
2675	1660		TAD I	DIAG2	
2676	3773		DCA	XPOINT	
2677	2260		ISZ	DIAG2	
2700	1660		TAD I	DIAG2	
2701	3772		DCA	YPOINT	
2702	5306		JMP	DIAG2A	
2703	6052		6052		/SKIP ON DONE.
2704	5303		JMP	1=1	/WAIT FOR DONE.
2705	6055		6055		/INTENSIFY
2706	1773	DIAG2A,	TAD	XPOINT	/GET X COORDINANT.
2707	6053		6053		/LOAD X
2710	7200		CLA		
2711	1772		TAD	YPOINT	/GET Y COORDINANT.
2712	6054		6054		/LOAD Y
2713	1371		TAD	1=1	/DECREMENT Y.
2714	3772		DCA	YPOINT	/SAVE Y.
2715	1773		TAD	XPOINT	/GET X COORDINANT.
2716	7001		IAC		/INCREMENT X.
2717	3773		DCA	XPOINT	/SAVE X.
2720	2257		ISZ	DIACNT	/IS PLOT COMPLETE?
2721	5303		JMP	DIAG2A=3	/NO, CONTINUE PLOT.
2722	2260		ISZ	DIAG2	/YES, SETUP RETURN.
2723	5660		JMP I	DIAG2	/RETURN
2771	7777				
2772	2241				
2773	2240				
2774	2435				
2775	2417				
2776	2462				
2777	2444				
	3000				

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//SUBROUTINES TO HANDLE OUTPUTS TO TTY77

3000	0000	CALSND, 0	
3001	4214	JMS	SIXTY
3002	0003	SUM1	
3003	3006	OUT1	
3004	4777	JMS	MASAGE
3005	3736	3736	
3006	7777	OUT1, 7777	
3007	7777	7777	
3010	4040	4040	
3011	0000	0000	
3012	6001	ION	
3013	5600	JMP I	CALSND
3014	0000	SIXTY, 0	
3015	7000	NOP	
3016	7000	NOP	
3017	7200	CLA	
3020	1614	TAD I	,+4
3021	3223	DCA	,+2
3022	5624	JMP I	,+2
3023	0000	0	
3024	3026	SIXTY+12	
3025	5217	JMP	SIXTY+3
3026	1623	TAD I	SIXTY+7
3027	0376	AND	(0007
3030	3271	DCA	MASKA
3031	1623	TAD I	SIXTY+7
3032	0375	AND	(0070
3033	3272	DCA	MASKB
3034	1623	TAD I	SIXTY+7
3035	0374	AND	(700
3036	3273	DCA	MASKC
3037	1623	TAD I	SIXTY+7
3040	0373	AND	(7000
3041	3274	DCA	MASKD
3042	1273	TAD	MASKC
3043	7112	RTR CLL	
3044	7010	RAR	
3045	1274	TAD	MASKD
3046	7012	RTR	
3047	7010	RAR	
3050	1275	TAD	MASKD+1
3051	3273	DCA	MASKC
3052	2214	ISZ	SIXTY
3053	4224	JMS	SIXTY+10
3054	1273	TAD	MASKC
3055	3623	DCA I	SIXTY+7
3056	1272	TAD	MASKB
3057	7004	RAL	
3060	7006	RTL	
3061	1271	TAD	MASKA
3062	1275	TAD	MASKD+1

3063	2223	ISZ	SIXTY+7
3064	3623	DCA I	SIXTY+7
3065	1372	TAD	(SIXTY+12
3066	3224	DCA	SIXTY+10
3067	2214	ISZ	SIXTY
3070	5614	JMP I	SIXTY
3071	0000	MASKA,	0
3072	0000	MASKB,	0
3073	0000	MASKC,	0
3074	0000	MASKD,	0
3075	6060		6060

3172	3026
3173	7000
3174	0700
3175	0070
3176	0007
3177	4660
	3200

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//MESSAGE LISTINGS//

3200	3736	MSG1,	TEXT	"*+VC8E POINT PLOT DISPLAY DIAGNOSTIC*"
3201	2603			
3202	7005			
3203	4020			
3204	1711			
3205	1624			
3206	4020			
3207	1417			
3210	2440			
3211	0411			
3212	2320			
3213	1401			
3214	3140			
3215	0411			
3216	0107			
3217	1617			
3220	2324			
3221	1103			
3222	3736			
3223	0000			

3224	3736	MSG2,	TEXT	"*+MAINDEC-8E-D6CA*"
3225	1501			
3226	1116			
3227	0405			
3230	0355			
3231	7005			
3232	5504			
3233	6603			
3234	0137			
3235	3600			

3236	3736	MSG3,	TEXT	"*+S,R,7=1, PERFORM TEST SELECTED BY S,R, 8-11*"
3237	2356			
3240	2256			
3241	6775			
3242	6154			
3243	4020			
3244	0522			
3245	0617			
3246	2215			
3247	4024			
3250	0523			
3251	2440			
3252	2305			
3253	1405			
3254	0324			
3255	0504			
3256	4002			
3257	3140			
3260	2356			
3261	2256			

3262 4070
3263 5561
3264 6137
3265 3600

3266 2356 MSG4, TEXT "S,R,7=0, RETURN TO DISPATCH ROUTINE TO GET NEXT TEST-"
3267 2256
3270 6775
3271 6054
3272 4022
3273 0524
3274 2522
3275 1640
3276 2417
3277 4004
3300 1123
3301 2001
3302 2403
3303 1040
3304 2217
3305 2524
3306 1116
3307 0540
3310 2417
3311 4007
3312 0524
3313 4016
3314 0530
3315 2440
3316 2405
3317 2324
3320 3736
3321 0000

3322 3736 MSG5, TEXT "←SELECT TEST←"
3323 2305
3324 1405
3325 0324
3326 4024
3327 0523
3330 2437
3331 3600

3332 3736 MSG6, TEXT "←DC CALIBRATION TEST←"
3333 0403
3334 4003
3335 0114
3336 1102
3337 2201
3340 2411
3341 1716
3342 4024
3343 0523
3344 2437
3345 3600

3346 3736 MSG7, TEXT "++RAMP TEST++"
3347 2201
3350 1520
3351 4024
3352 0523
3353 2437
3354 3600

3355 3736 MSG8, TEXT "++CONTROL LOGIC TEST++"
3356 0317
3357 1624
3360 2217
3361 1440
3362 1417
3363 0711
3364 0340
3365 2405
3366 2324
3367 3736
3370 0000

3371 3736 MSG9, TEXT "++VERTICAL FLYBACK++"
3372 2605
3373 2224
3374 1103
3375 0114
3376 4006
3377 1431
3400 0201
3401 0313
3402 3736
3403 0000

3404 3736 MSG10, TEXT "++HORIZONTAL FLYBACK++"
3405 1017
3406 2211
3407 3217
3410 1624
3411 0114
3412 4006
3413 1431
3414 0201
3415 0313
3416 3736
3417 0000

3420 3736 MSG11, TEXT "++CORNERS TEST++"
3421 0317
3422 2216
3423 0522
3424 2340
3425 2405
3426 2324
3427 3736
3430 0000

3431	3736	MSG12, TEXT	"*VERTICAL BAR TEST*"
3432	2605		
3433	2224		
3434	1103		
3435	0114		
3436	4002		
3437	0122		
3440	4024		
3441	0523		
3442	2437		
3443	3600		

3444	3736	MSG13, TEXT	"*HORIZONTAL BAR TEST*"
3445	1017		
3446	2211		
3447	3217		
3450	1624		
3451	0114		
3452	4002		
3453	0122		
3454	4024		
3455	0523		
3456	2437		
3457	3600		

3460	3736	MSG14, TEXT	"*SINGLE POINT PLOT TEST*"
3461	2311		
3462	1607		
3463	1405		
3464	4020		
3465	1711		
3466	1624		
3467	4020		
3470	1417		
3471	2440		
3472	2405		
3473	2324		
3474	3736		
3475	0000		

3476	3736	MSG15, TEXT	"*PUT DESIRED VALUE OF X IN S,R, AND PRESS CONTINUE*"
3477	2025		
3500	2440		
3501	0405		
3502	2311		
3503	2205		
3504	0440		
3505	2601		
3506	1425		
3507	0540		
3510	1706		
3511	4030		
3512	4011		
3513	1640		

3514 2356
3515 2256
3516 4001
3517 1604
3520 4020
3521 2205
3522 2323
3523 4003
3524 1716
3525 2411
3526 1625
3527 0537
3530 3600

3531 3736 MSG16, TEXT "++PUT DESIRED VALUE OF Y IN S.R. AND PRESS CONTINUE++"

3532 2025
3533 2440
3534 0405
3535 2311
3536 2205
3537 0440
3540 2601
3541 1425
3542 0540
3543 1706
3544 4031
3545 4011
3546 1640
3547 2356
3550 2256
3551 4001
3552 1604
3553 4020
3554 2205
3555 2323
3556 4003
3557 1716
3560 2411
3561 1625
3562 0537
3563 3600

3564 3736 MSG17, TEXT "++SET SW7=1 TO LOOP IN TEST"

3565 2305
3566 2440
3567 2327
3570 6775
3571 6140
3572 2417
3573 4014
3574 1717
3575 2040
3576 1116
3577 4024
3600 0523

3601 2400

3602 3736

3603 2305

3604 2440

3605 2327

3606 6775

3607 6040

3610 2417

3611 4020

3612 1417

3613 2440

3614 2017

3615 1116

3616 2440

3617 1716

3620 0305

3621 3440

3622 2022

3623 0523

3624 2340

3625 0317

3626 1624

3627 1116

3630 2505

3631 3736

3632 0000

3633 3736

3634 0411

3635 0107

3636 1716

3637 0114

3640 2337

3641 3600

3642 3736

3643 0322

3644 1723

3645 2311

3646 1607

3647 4004

3650 1101

3651 0717

3652 1601

3653 1423

3654 3736

3655 0000

3656 3736

3657 0411

3660 2320

3661 1401

3662 3105

3663 0440

MSG18, TEXT

"++SET SW7=0 TO PLOT POINT ONCE, PRESS CONTINUE++"

MSG19, TEXT

"++DIAGONALS++"

MSG20, TEXT

"++CROSSING DIAGONALS++"

MSG21, TEXT

"++DISPLAYED CALIBRATION TEST++"

3664	0301
3665	1411
3666	0222
3667	0124
3670	1117
3671	1640
3672	2405
3673	2324
3674	3736
3675	0000

//CONTROL LOGIC ERROR MESSAGES//

3676	3736	EMSG1, TEXT	"*CLT1-DILE FAILED TO CLEAR AC*"
3677	0314		
3700	2461		
3701	5504		
3702	1114		
3703	0540		
3704	0601		
3705	1114		
3706	0504		
3707	4024		
3710	1740		
3711	0314		
3712	0501		
3713	2240		
3714	0103		
3715	3736		
3716	0000		

3717	3736	EMSG1A, TEXT	"*CLT1A-CHANNEL F/F NOT SET OR NOT READ BACK*"
3720	0314		
3721	2461		
3722	0155		
3723	0310		
3724	0116		
3725	1605		
3726	1440		
3727	0657		
3730	0640		
3731	1617		
3732	2440		
3733	2305		
3734	2440		
3735	1722		
3736	4016		
3737	1724		
3740	4022		
3741	0501		
3742	0440		
3743	0201		
3744	0313		
3745	3736		
3746	0000		

3747	3736	EMSG1B, TEXT	"*CLT1B-INTERRUPT ENABLE NOT SET OR NOT READ BACK*"
3750	0314		
3751	2461		
3752	0255		
3753	1116		
3754	2405		
3755	2222		
3756	2520		
3757	2440		

3760 0516
3761 0102
3762 1405
3763 4016
3764 1724
3765 4023
3766 0524
3767 4017
3770 2240
3771 1617
3772 2440
3773 2205
3774 0104
3775 4002
3776 0103
3777 1337
4000 3600

4001 3736
4002 0314
4003 2461
4004 0355
4005 0310
4006 0116
4007 1605
4010 1440
4011 0657
4012 0640
4013 1617
4014 2440
4015 2305
4016 2440
4017 1722
4020 4016
4021 1724
4022 4022
4023 0501
4024 0440
4025 0201
4026 0313
4027 3736
4030 0000

EMSG1, TEXT

"CLT1C-CHANNEL F/F NOT SET OR NOT READ BACK"

4031 3736
4032 0314
4033 2462
4034 5504
4035 1114
4036 0540
4037 0601
4040 1114
4041 0504
4042 4024
4043 1740
4044 0314

EMSG2, TEXT

"CLT2-DILE FAILED TO CLEAR ENABLE REGISTER"

4045 0501
4046 2240
4047 0516
4050 0102
4051 1405
4052 4022
4053 0507
4054 1123
4055 2405
4056 2237
4057 3600

4060 3736
4061 0314
4062 2463
4063 5504
4064 1114
4065 3040
4066 0601
4067 1114
4070 0504
4071 4024
4072 1740
4073 2305
4074 2440
4075 0417
4076 1605
4077 4017
4100 2240
4101 0314
4102 0501
4103 2205
4104 0440
4105 0103
4106 3736
4107 0000

EMSG3, TEXT "CLT3-DILX FAILED TO SET DONE OR CLEARED AC"

4110 3736
4111 0314
4112 2464
4113 5504
4114 1114
4115 3140
4116 0601
4117 1114
4120 0504
4121 4024
4122 1740
4123 2305
4124 2440
4125 0417
4126 1605
4127 4017
4130 2240
4131 0314

EMSG4, TEXT "CLT4-DILY FAILED TO SET DONE OR CLEARED AC"

4132 0501
4133 2205
4134 0440
4135 0103
4136 3736
4137 0000

4140 3736 MSG5, TEXT "CLT5=DIXY FAILED TO SET DONE"

4141 0314
4142 2465
4143 5504
4144 1130
4145 3140
4146 0601
4147 1114
4150 0504
4151 4024
4152 1740
4153 2305
4154 2440
4155 0417
4156 1605
4157 3736
4160 0000

4161 3736 MSG6, TEXT "CLT6=DICD FAILED TO CLEAR DONE FLAG"

4162 0314
4163 2466
4164 5504
4165 1103
4166 0440
4167 0601
4170 1114
4171 0504
4172 4024
4173 1740
4174 0314
4175 0501
4176 2240
4177 0417
4200 1605
4201 4006
4202 1401
4203 0737
4204 3600

4205 3736 MSG7, TEXT "CLT7=DISD FAILED TO SKIP ON DONE FLAG OR CLRD FLG"

4206 0314
4207 2467
4210 5504
4211 1123
4212 0440
4213 0601
4214 1114
4215 0504

4216 4024
4217 1740
4220 2313
4221 1120
4222 4017
4223 1640
4224 0417
4225 1605
4226 4006
4227 1401
4230 0740
4231 1722
4232 4003
4233 1422
4234 0440
4235 0614
4236 0737
4237 3600

MSG8, TEXT "CLT8=ILLEGAL INT, CAUSED BY INT, ENABLER"

4240 3736
4241 0314
4242 2470
4243 5511
4244 1414
4245 0507
4246 0114
4247 4011
4250 1624
4251 5640
4252 0301
4253 2523
4254 0504
4255 4002
4256 3140
4257 1116
4260 2456
4261 4005
4262 1601
4263 0214
4264 0537
4265 3600

MSG9, TEXT "CLT9=ILLEGAL INT, CAUSED BY DONE FLAG"

4266 3736
4267 0314
4270 2471
4271 5511
4272 1414
4273 0507
4274 0114
4275 4011
4276 1624
4277 5640
4300 0301
4301 2523
4302 0504

4303 4002
4304 3140
4305 0417
4306 1605
4307 4006
4310 1401
4311 0737
4312 3600

4313 3736 MSG10, TEXT "++CLT10=INT. ENABLE AND DONE FLG DID NOT INTERRUPT++"
4314 0314
4315 2461
4316 6055
4317 1116
4320 2456
4321 4005
4322 1601
4323 0214
4324 0540
4325 0116
4326 0440
4327 0417
4330 1605
4331 4006
4332 1407
4333 4004
4334 1104
4335 4016
4336 1724
4337 4011
4340 1624
4341 0522
4342 2225
4343 2024
4344 3736
4345 0000

4346 3736 MSG11, TEXT "++CLT11=DILX FAILED TO CLEAR DONE++"
4347 0314
4350 2461
4351 6155
4352 0411
4353 1430
4354 4006
4355 0111
4356 1405
4357 0440
4360 2417
4361 4003
4362 1405
4363 0122
4364 4004
4365 1716
4366 0537
4367 3600

4370 3736 MSG12, TEXT

"CLY12=DILY FAILED TO CLEAR DONE"

4371 0314

4372 2461

4373 6255

4374 0411

4375 1431

4376 4006

4377 0111

4400 1405

4401 0440

4402 2417

4403 4003

4404 1405

4405 0122

4406 4004

4407 1716

4410 0537

4411 3600

4600

PAGE

/MESSAGE TYPE=OUT ROUTINE

4600	0000	MESSAGE, 0	
4601	7240	CLA CMA	
4602	1600	TAD I	MESSAGE
4603	3010	DCA	10
4604	2200	ISZ	MESSAGE
4605	1410	TAD I	10
4606	3217	DCA	MSRGHT
4607	1217	TAD	MSRGHT
4610	7012	RTR	
4611	7012	RTR	
4612	7012	RTR	
4613	4220	JMS	TYPECH
4614	1217	TAD	MSRGHT
4615	4220	JMS	TYPECH
4616	5205	JMP	MESSAGE+5
4617	0000	MSRGHT, 0	
4620	0000	TYPECH, 0	
4621	0252	AND	MASK77
4622	7450	SNA	
4623	5600	JMP I	MESSAGE
4624	1253	TAD	M40
4625	7510	SPA	
4626	5231	JMP	,+3
4627	1254	TAD	C240
4630	5244	JMP	MTP
4631	7001	IAC	
4632	7440	SZA	
4633	5236	JMP	,+3
4634	1255	TAD	C215
4635	5244	JMP	MTP
4636	7001	IAC	
4637	7440	SZA	
4640	5243	JMP	,+3
4641	1256	TAD	C212
4642	5244	JMP	MTP
4643	1257	TAD	C336
4644	6046	MTP, TLS	
4645	6041	TSF	
4646	5245	JMP	,=1
4647	6042	TCF	
4650	7200	CLA	
4651	5620	JMP I	TYPECH
4652	0077	MASK77, 77	
4653	7740	M40, -40	
4654	0240	C240, 240	
4655	0215	C215, 215	
4656	0212	C212, 212	
4657	0336	C336, 336	

//MESSAGE ROUTINE FOR CALIBRATION VALUES//

```

4660 0000  MASAGE, 0
4661 6002  IOF
4662 7240  CLA CMA
4663 1260  TAD  MASAGE
4664 3010  DCA  10
4665 1410  TAD I  10
4666 3277  DCA  MSRG
4667 1277  TAD  MSRG
4670 7012  RTR
4671 7012  RTR
4672 7012  RTR
4673 4300  JMS  TYPEC
4674 1277  TAD  MSRG
4675 4300  JMS  TYPEC
4676 5265  JMP  MASAGE+5
4677 0000  MSRG, 0
4700 0000  TYPEC, 0
4701 0252  AND  MASK77
4702 7450  SNA
4703 5410  JMP I  10
4704 1253  TAD  M40
4705 7510  SPA
4706 5311  JMP  ,+3
4707 1254  TAD  C240
4710 5324  JMP  MTPA
4711 7001  IAC
4712 7440  SEA
4713 5316  JMP  ,+3
4714 1255  TAD  C215
4715 5324  JMP  MTPA
4716 7001  IAC
4717 7440  SEA
4720 5323  JMP  ,+3
4721 1256  TAD  C212
4722 5324  JMP  MTPA
4723 1257  TAD  C336
4724 6046  MTPA, TLS
4725 6041  TSF
4726 5325  JMP  ,+1
4727 6042  TCF
4730 7200  CLA
4731 5700  JMP I  TYPEC

```

\$

```

0171 4600
0172 0200
0173 7777
0174 0002
0175 0400
0176 0020
0177 5000

```

[illegible]

4000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4400	11111111	11000000	00000000	00000000	00000000	00000000	00000000	00000000
4500	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
4600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
4700	11111111	11111111	11111111	11000000	00000000	00000000	00000000	00000000

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

C212	4656	DISP2	2462	MSG12	3431	SIXTY	3014
C215	4655	DISP3	2417	MSG13	3444	START	0200
C240	4654	DISP4	2435	MSG14	3460	SUM1	0003
C336	4657	DISPAT	0216	MSG15	3476	SWITCH	0455
CAF	6007	DISRET	0020	MSG16	3531	TABLE	0434
CALCNT	0456	DISTST	2242	MSG17	3564	TABLEA	2313
CALIB	0403	DIXY	4155	MSG18	3602	TALLY	0005
CALSND	3000	EMSG1	3676	MSG19	3633	TALLYA	0021
CALTST	0400	EMSG10	4313	MSG2	3224	TEST	0234
CALX	0415	EMSG11	4346	MSG20	3642	TITLE	0255
CALY	0412	EMSG12	4370	MSG21	3656	TYPEC	4700
CKSW7	0036	EMSG1A	3717	MSG3	3236	TYPECH	4620
CLT1	0611	EMSG1B	3747	MSG4	3266	UPDVAL	0421
CLT10	1221	EMSG1C	4001	MSG5	3322	VARYCT	2272
CLT11	1240	EMSG2	4031	MSG6	3332	VERBAR	2036
CLT11A	1250	EMSG3	4060	MSG7	3346	VERCNT	2111
CLT12	1243	EMSG4	4110	MSG8	3355	VERFLY	1441
CLT12A	1273	EMSG5	4140	MSG9	3371	VERTST	1434
CLT1A	0624	EMSG6	4161	MSGPNT	0035	VR0YST	2023
CLT1B	0641	EMSG7	4205	MSRGHT	4617	XCROS	2404
CLT1C	0656	EMSG8	4240	MSRGT	4677	XD1CD	0134
CLT2	0673	EMSG9	4266	MTP	4644	XD1CL	0130
CLT3	0710	EOUT	0126	MTPA	4724	XD1LE	0161
CLT4	1001	ERR8	1126	NXTST	0022	XD1LX	0145
CLT5	1035	ERR9	1213	OK10	1235	XD1LY	0151
CLT6	1051	ERRMSG	0062	OUT1	3006	XD1RE	0165
CLT7	1065	ERROR	4101	PLINE	2443	XD1SD	0140
CLT8	1114	ERSWIT	0004	PLOTCT	2470	XD1XY	0155
CLT9	1201	ERTYPE	0122	PNTR1	0497	XPOINT	2240
CLTST	0600	FAIL	0101	PNTR2	2326	XRAMP	0473
CORNER	1604	GETBAK	0023	PNTST	2200	XVERT	2056
CORTST	1600	GETVAL	0417	PRESET	0025	XX	7402
CROTST	2400	HORBAR	2072	RAMP	0463	YPOINT	2241
DBTST	2000	HORCNT	2055	RAMPA	0476	YVERT	2112
DELAY	0024	HORFLY	1405	RD1CD	0135		
DIABIS	2004	HORTST	1400	RD1CL	0131		
DIACNT	2657	HRBTST	2057	RD1LE	0162		
DIAG1	2626	IOTY	0056	RD1LX	0146		
DIAG1A	2650	K7077	0320	RD1LY	0192		
DIAG2	2660	M40	4653	RD1RE	0166		
DIAG2A	2706	MASAGE	4660	RD1SD	0141		
DICD	4134	MASK77	4652	RD1XY	0156		
DICL	4130	MASKA	3071	RETURN	0002		
DILE	4161	MASKB	3072	RMPTST	0460		
DILX	4145	MASKC	3073	SAVIT	0504		
DILY	4151	MASKD	3074	SEL1	0053		
DIRE	4165	MESAGE	4600	SELCHN	0044		
DISCAL	2245	MIOT	0321	SETHOR	2613		
DISD	4140	MLINE	2416	SETIOT	0256		
DISLOP	2265	MSG1	3200	SETUP	4025		
DISMSG	0213	MSG10	3404	SETVER	2600		
DISP1	2444	MSG11	3420	SINPNT	2222		

ERRORS DETECTED: 0

LINKS GENERATED: 85

RUN-TIME: 14 SECONDS

3K CORE USED